

Domains

A: Mental-health journaling (anchor)

AI Conditions: C0, C1, C2, C3

Seeds per condition: 3 (A), 2 (B/C)

Human baselines: HB1 (FG), HB2 (EGM)

C1: 3-step; C2: A→B; C3: A→B→C

B: Personal Finance/budgeting (satellite)

AI Conditions: C0, C1, C3

Seeds per condition: 2

Human baselines: HB1, HB2

C: Fitness/habit tracking (satellite)

AI Conditions: C0, C1, C3

Seeds per condition: 2

Human baselines: HB1, HB2

AI run order randomized within domain.

Randomization & Counterbalancing
HB1→HB2 vs HB2→HB1; ≥1-week washout.
Pre-registered seeds; full logging

S1 Persona generation



C0: one-shot (6 personas)

C1–C3: 3-step (2→4→6 personas)

Creator–voice mapping enforced

S2 AI focus groups



6 personas; ≤50–60 turns/session

Guardrails: safety, role integrity, logging

≤2 challenge turns after Q2 if conflict is weak

Data process

- Requirement extraction & deduplication
- Normalize; embed+cluster; coder adjudication
- Default threshold: cosine ≥ 0.82

- Blind cross-ratings
- Raters: relevance, clarity, feasibility, innovation
- Validation rule: median ≥ 4 or ≥ 60% ≥ 4; source-blind

S3 Post-validation reflective interviews



Explain why items resonate; hybrid guidance

Validation is blind; reflective interviews follow validation (source revealed only after initial judgments).

Human Baselines

HB1: Human focus group (FG)

N=6 per domain; same five surfaces; moderator
Transcripts coded; requirements extracted

HB2: Emotional Goal Modeling (EGM)

Outputs: goal graph, traced requirements
N=6 per domain;
goal→subgoal→requirement tracing



Outputs

(recorded for analysis & reproducibility)



Persona diversity

TTR, Attribute diversity,
Semantic dispersion



Requirement quality

Relevance, clarity,
feasibility, innovation

AI Conditions

C0: 1-LLM one-shot

C1: 1-LLM, 3-step (2→4→6)

C2: 2-LLM, A→B across steps

C3: 3-LLM, A→B→C

C2 appears only in Domain A